

8-BIT ARITHMETIC LOGIC UNIT (ALU)

DESIGN SPECIFICATION

1. Overview

An 8-bit combinational ALU that performs standard arithmetic, logical, and bitwise operations based on a 3-bit input opcode.

2. Signal Descriptions

Signal	Direction / Width	Description
A	Input, 8-bit	Primary operand A
B	Input, 8-bit	Secondary operand B
opcode	Input, 3-bit	Operation select code
Result	Output, 8-bit	Output result
carryout	Output, 1-bit	High on addition overflow / subtraction underflow
zero_flag	Output, 1-bit	High if Result equals 0x00

3. Supported Operations

Opcode	Operation	Function
3'b000	Pass-Through	Result = A
3'b001	Addition	{carryout, Result} = A + B
3'b010	Subtraction	{carryout, Result} = A - B
3'b011	Increment	Result = A + 1
3'b100	Decrement	Result = A - 1
3'b101	Bitwise AND	Result = A & B
3'b110	Bitwise OR	Result = A B
3'b111	Bitwise NOT	Result = ~A

4. Status Flags

- Carry Flag is active only for addition and subtraction operations.
- Zero Flag is driven asynchronously based on $\text{Result} == 0$.